

# Adaptive Inference for Functionals of M-Estimands



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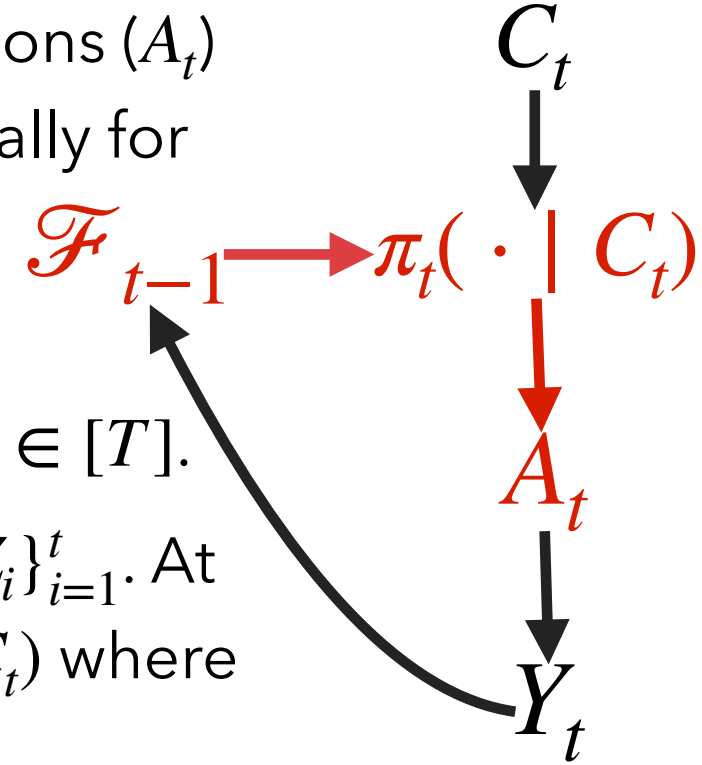


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## Inference for Contextual Bandits

**Problem Setup** Assume features ( $C_t$ ), actions ( $A_t$ ) and outcomes ( $Y_t$ ) are observed sequentially for  $T$  rounds. Let  $Z_t := (C_t, A_t, Y_t)$ .

- $\{C_t\}_{t=1}^T \sim P_C$  are i.i.d. and  $Y_t | C_t, A_t \sim P_Y(\cdot | C_t, A_t)$  is fixed across  $t \in [T]$ .
- Let  $\mathcal{F}_t$  be the filtration generated by  $\{Z_i\}_{i=1}^t$ . At each  $t$ , the analyst samples  $A_t \sim \pi_t(\cdot | C_t)$  where  $\pi_t \in \mathcal{F}_{t-1}$ .



**Prior work** notes that when the variance does not stabilize to a deterministic limit, asymptotic normality of classical estimators (e.g. OLS) may not hold (Zhang et al., 2020). Adjustments have been proposed for linear regression (e.g. Deshpande et al., 2018; Khamaru et al., 2025), partial linear regression (e.g. Lin et al., 2025), and  $M$ -estimators (e.g. Zhang et al., 2021).

### Our work

- Allows for flexible nonparametric estimation of potentially infinite dimensional nuisance parameters
- Allows for incorrectly specified working models
- Many settings studied in prior work are special cases of the general framework

**Inferential Target** Choose some fixed policy  $\pi_{e^*}$ , loss function  $\ell$ , nuisance  $\eta_{P_e}$  and evaluation function  $m$ . Define

$$\theta_{P_e} := \operatorname{argmin}_{\theta \in \Theta} \mathbb{E}_{P_e} [\ell_{\theta, \eta_{P_e}}(Z)]$$

$$\Psi_e(\theta_{P_e}) := \mathbb{E}_{P_e} [m_{\theta_{P_e}}(Z)]$$

Note that  $\theta_{P_e}$  and  $\eta_{P_e}$  may be infinite dimensional

### Example

|                           | $\theta_{P_e}$                                                                         | $\Psi_e(\theta_{P_e})$                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| ATE                       | $\mathbb{E}[Y   A = a, C = c]$                                                         | $\mathbb{E}_{P_C} [\theta_{P_e}(1, C) - \theta_{P_e}(0, C)]$                                              |
| M-estimator (parametric)  | $\operatorname{argmin}_{\theta \in \Theta} \mathbb{E}_{P_e} [\ell_{\theta}(C, A, Y)]$  | $v^\top \theta_{P_e}$ , fixed $v$                                                                         |
| Partial linear regression | $\operatorname{argmin}_{\theta} \mathbb{E}_{P_e} [(Y - \theta^\top X - \eta_{P_e})^2]$ | $v^\top \theta_{P_e}$ , fixed $v$                                                                         |
| Value of optimal policy   | $\mathbb{E}[Y   A = a, C = c]$                                                         | $\sup_{\pi \in \Pi} \mathbb{E}_P \left[ \int_{\mathcal{A}} \pi(a   C) \theta_{P_e}(a, C) \mu(da) \right]$ |

## Asymptotic Normality

In the nonparametric model and under suitable regularity conditions, the canonical gradient at the evaluation law where  $A \sim \pi_e$  is given by

$$\varphi_{P_e}(Z) := m_{\theta_{P_e}}(Z) - \Psi(P_e) - \partial_{\theta} \ell_{\theta_{P_e}, \eta_{P_e}}(Z) [\alpha_{P_e}]$$

$\alpha_{P_e}$  denotes the Riesz representer that equates

$$E_{P_e} [\partial_{\theta} m_{\theta_{P_e}}(Z) [h]] = \mathbb{E}_{P_e} [\partial_{\theta}^2 \ell_{\theta_{P_e}, \eta_{P_e}}(Z) [h, \alpha_{P_e}]] \text{ for all } h \in \Theta.$$

**Key Fact:** If  $w_t := \frac{\pi_e(A_t | C_t)}{\pi_t(A_t | C_t)}$  and  $\varphi_t = \varphi_{P_e}(Z_t)$  then  $\{w_t \varphi_t\}_{t=1}^T$  is a martingale difference sequence because  $\mathbb{E} [w_t \varphi_t | \mathcal{F}_{t-1}] = \mathbb{E}_{P_e} [\varphi_{P_e}(Z)] = 0$ .

Does  $\sigma_t^2 := \mathbb{E}[w_t^2 \varphi_t^2 | \mathcal{F}_{t-1}]$  converge to an a.s. finite random variable  $\kappa$ ?

**Yes**  $\rightarrow$  Use self-normalized CLT  
**No**  $\rightarrow$  Use stabilized MDS  $\{\sigma_t^{-1} w_t \varphi_t\}_{t=1}^T$

### Stabilized one-step estimator (Theorem 2; simplified)

Use when  $\sigma_t$  **does not** stabilize.

Assume estimates  $(\hat{\theta}_t, \hat{\eta}_t, \hat{\alpha}_t)$  are  $\mathcal{F}_{t-1}$  measurable and converge at  $O_p(t^{-1/4})$  rates. Assume  $\hat{\sigma}_t$  is  $\mathcal{F}_{t-1}$  measurable and consistently tracks  $\sigma_t$ .

$$\hat{\Psi}_T^{\text{os}} := \left( \sum_{t=1}^T \hat{\sigma}_t^{-1} w_t \right)^{-1} \sum_{t=1}^T \hat{\sigma}_t^{-1} w_t \left( m_{\hat{\theta}_t}(Z_t) - \partial_{\theta} \ell_{\hat{\theta}_t, \hat{\eta}_t}(Z_t) [\hat{\alpha}_t] \right)$$

Under appropriate regularity conditions (e.g. conditional Lindeberg, policy overlap)

$$T^{-1/2} \left( \sum_{t=1}^T \hat{\sigma}_t^{-1} w_t \right) (\hat{\Psi}_T^{\text{os}} - \Psi_e(\theta_{P_e})) \xrightarrow{d} N(0, 1)$$

### Self-normalized CLT (Theorem 3; simplified)

Use when  $\sigma_t$  **converges to a random limit**.

Avoid estimating  $\sigma_t$  and use realized quadratic variation. Define

$$\hat{V}_T := \left( \sum_{t=1}^T w_t \right)^{-2} \sum_{t=1}^T w_t^2 \hat{\varphi}_t^2 \quad (\text{where } \hat{\varphi}_t \text{ is a consistent estimate of the IF})$$

Then under similar regularity conditions as above,

$$\hat{V}_T^{-1/2} \left( \sum_{t=1}^T w_t \right)^{-1} \left( \sum_{t=1}^T w_t \left( m_{\hat{\theta}_t}(Z_t) - \partial_{\theta} \ell_{\hat{\theta}_t, \hat{\eta}_t}(Z_t) [\hat{\alpha}_t] \right) - \Psi_e(\theta_{P_e}) \right) \xrightarrow{d} N(0, 1)$$

### Conditional Variance Estimator (plug-in for Theorem 2)

$$\hat{\sigma}_t^2 := \frac{1}{t-1} \sum_{s=1}^{t-1} \frac{\pi_e^2(A_s | C_s)}{\pi_t(A_s | C_s) \pi_s(A_s | C_s)} \hat{\varphi}_s^2(Z_s)$$

**Requirements** The policies  $\{\pi_t\}_{t=1}^T$  must lie in a class  $\Pi$  such that

- $\sup_{\pi \in \Pi} \sup_{c, a \in \mathcal{C} \times \mathcal{A}} \frac{\pi_e(a | c)}{\pi(a | c)} = O(T^{1/8})$
- $\log N(\epsilon, \log \Pi, \|\cdot\|_{\infty}) \lesssim \begin{cases} \log(e/\epsilon), & p = 0, \\ \epsilon^{-p}, & p \in (0, 2), \end{cases}$

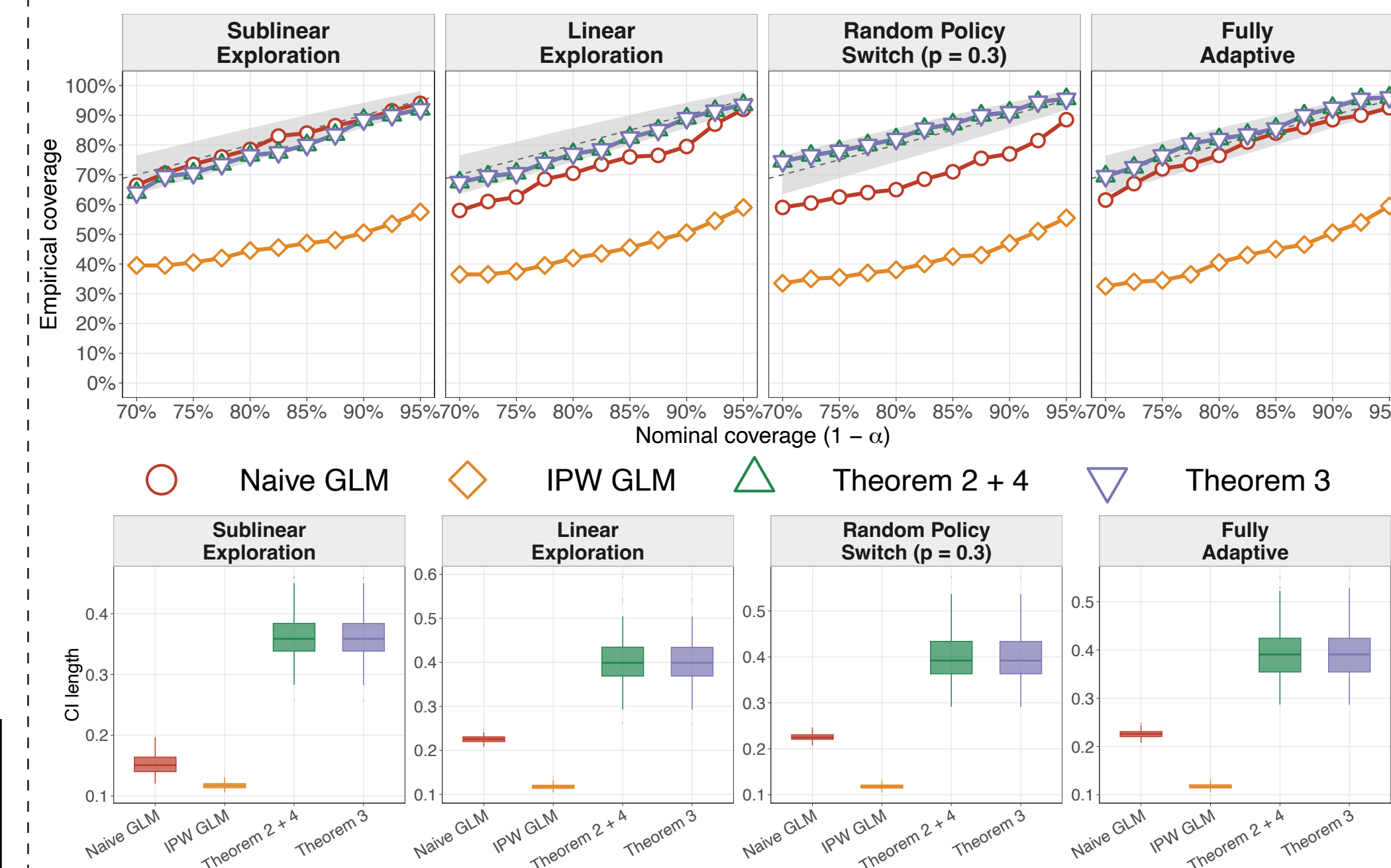
## Simulations

We model customer renewal decisions  $Y_t \in \{0, 1\}$  as a function of contexts and prices ( $p_t$ ) follow a logistic model

$$\operatorname{logit} \mathbb{P}(Y_t = 1 | C_t, p_t) = g^* + \tau g_0(C_t) - \beta p_t$$

At each  $t \in [T]$ , the statistician attempts to maximize revenue

$$R(p) := \mathbb{E}_{P_C} [p \cdot \Lambda(g^* + \tau g_0(C) - \beta p)]$$



We track: policies that explore prices for either **sublinear** or **linear** number of rounds before commitment, policies that change pricing at random times, and policies that adjust prices at every round.

After the experiment, we attempt to construct CIs covering  $\beta$

CIs constructed with our methods remain valid but using out-of-the-box GLMs and IPWs undercover in **all settings**.